

Pranav Varshney

Bengaluru, India

prnvvarshney@gmail.com [LinkedIn](#) [Portfolio](#) [GitHub](#)

Professional Summary

Full-stack Software Engineer delivering production systems at Olympic scale — building real-time sports and Olympics management platforms for NBCUniversal serving millions of concurrent users. Drives measurable impact through Angular + .NET engineering, high-throughput API design, and cloud-native DevOps. Microsoft Azure Developer Associate certified, with a hands-on track record of cutting latency, accelerating deployments, and shipping reliably under live-event pressure.

Technical Skills

Languages: C#, Java, JavaScript, TypeScript, HTML, CSS

Frameworks: .NET Core, ASP.NET Web API, Angular, Node.js, REST APIs, WebSockets, WebRTC

Cloud & DevOps: AWS, Azure, Docker, Jenkins, GitHub Actions, Proxmox, CI/CD

Databases: MySQL, MongoDB

Tools: Git, Grafana, Figma, Adobe Suite

Professional Experience

Software Engineer – Capgemini

Dec 2024 – Present

Client: NBCUniversal Comcast | Bengaluru, India

- Designed and delivered Angular + .NET modules for real-time sports and Olympics management platforms, owning end-to-end feature development from architecture to production deployment under live-event traffic for millions of concurrent users.
- Identified and resolved backend API and database bottlenecks, reducing average response time by ~35% under peak broadcast load — directly improving platform reliability during high-stakes events.
- Proposed and built a real-time data synchronization engine for live event tracking, cutting data-update latency by ~40% through efficient state management and WebSocket-based data flow.
- Redesigned CI/CD pipelines using Jenkins and GitHub Actions, cutting average deployment time by 30% and eliminating error-prone manual release steps.
- Led cross-functional collaboration across multiple time zones in an agile environment, consistently delivering sprint commitments on schedule with zero missed releases.

Full Stack Developer Intern – Bharat Intern

Jul 2023 – Aug 2023

Remote

- Built a multi-user real-time video conferencing platform using Angular, Node.js, and WebRTC, achieving stable audio/video streaming for up to 10 concurrent participants.
- Designed and implemented secure RESTful backend APIs with session-based authentication and MySQL-backed user management.
- Diagnosed and resolved WebRTC signaling bottlenecks, reducing end-to-end stream latency through optimized connection negotiation and ICE candidate handling.

Projects

AgriTrade – Farmer to Dealer Marketplace Platform

(.NET Core, Angular, MySQL)

- Architected a full-stack B2B marketplace enabling farmers and dealers to trade crops through role-based dashboards, streamlining the agricultural supply chain by eliminating manual negotiation steps.
- Designed JWT-secured REST APIs covering authentication, product listing, and order workflows, with input validation and role-based access control.
- Optimized MySQL queries and indexing strategy, cutting average inventory query time by ~50% on datasets of 10K+ records.

Real-Time Video Conferencing Application

(Angular, Node.js, WebRTC, MySQL)

- Built a peer-to-peer multi-user conferencing system with real-time audio, video, and text chat using WebRTC, achieving sub-200ms end-to-end latency under normal network conditions.

- Implemented a custom signaling server in Node.js to manage peer connections, ICE candidate exchange, and session state across multiple concurrent rooms.
- Developed secure backend services for user authentication, session persistence, and room management backed by MySQL.

Private Cloud Deployment using Proxmox

(Proxmox VE, Virtualization, Linux Networking)

- Designed and deployed a private cloud infrastructure on Proxmox VE, provisioning 5+ virtual machines with dedicated storage pools and snapshot-based backup policies.
- Configured isolated virtual networks with VLAN segmentation and firewall rules, hardening inter-VM security and reducing attack surface.
- Implemented Grafana-based monitoring dashboards and role-based access control, sustaining 99%+ uptime across a 3-month self-hosted production period.

Creative & UI Projects

- **PacMan Game** – Browser-based game built from scratch in JavaScript featuring tile-based collision detection, ghost AI, and score tracking — demonstrates low-level game loop and state management proficiency. [\[Live\]](#) [\[Code\]](#)
- **3D Particle System** – Real-time GPU-accelerated particle simulation with configurable motion physics and dynamic lighting, built with WebGL — showcases graphics programming and shader optimization skills. [\[Live\]](#) [\[Code\]](#)
- **3D Scene Environment** – Interactive WebGL scene with real-time object transformations, Phong shading, and free-camera navigation — highlights 3D rendering pipeline knowledge. [\[Live\]](#) [\[Code\]](#)
- **Portfolio Website** – Personal site showcasing UI/UX work and interactive web experiments, designed entirely in Figma and implemented with custom CSS animations. [\[Live\]](#)

Education

B.Tech in Computer Science Engineering (Cloud Computing & Virtualization)

GLA University, Mathura

2020 – 2024

Certifications

Microsoft Certified: Azure Developer Associate (2025)
 Generative AI Specialist – DeepLearning.AI (2025)
 IBM Cloud Computing & Virtualization Graduate (2024)
 Practical DevSecOps with GitHub Actions – EC-Council